

Prolog Program to Suggest a Dieting System Based on Disease

1. Aim

To write and execute a Prolog program that suggests an appropriate diet plan for a person based on their diagnosed disease, using facts, rules, and queries.

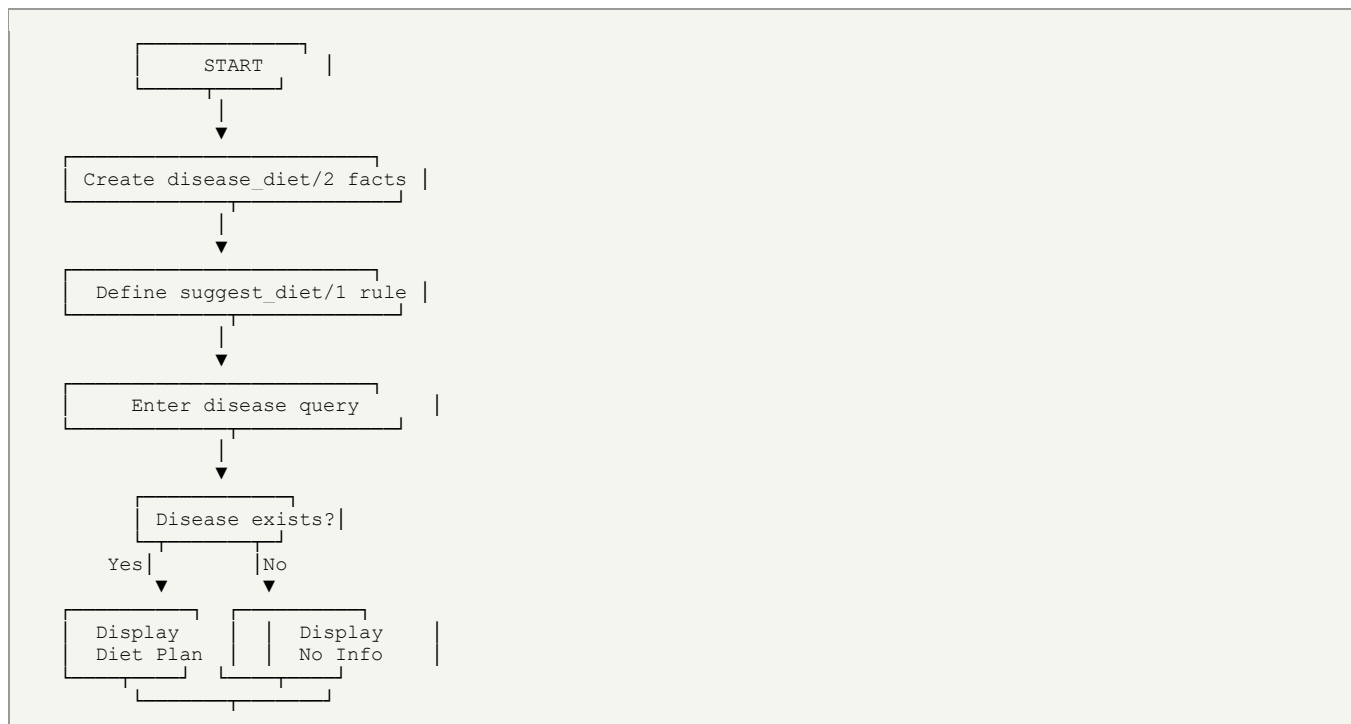
2. Objective

- To understand how expert-system-style advice can be modelled in Prolog.
- To represent disease-to-diet mappings as facts.
- To retrieve a suitable diet recommendation using a rule.
- To handle the case where no diet information is available for a disease.

3. Algorithm

1. Start the program.
2. Create a Prolog database using the predicate `disease_diet/2`.
3. Store each disease along with its recommended diet as a fact.
4. Define a rule `suggest_diet/1` to look up and display the diet for a disease.
5. Enter the disease name as a query.
6. Search the database for a matching disease.
7. If found, display the recommended diet; otherwise display that no information is available.
8. Stop the program.

4. Flowchart



END

5. Code with Output

Code:

```
% -----  
% Disease - Diet Database  
% -----  
disease_diet(diabetes,      'Low sugar, high fiber diet').  
disease_diet(hypertension,  'Low sodium diet').  
disease_diet(obesity,       'Low calorie, high protein diet').  
disease_diet(anemia,        'Iron rich diet').  
disease_diet(heart_disease, 'Low fat, low cholesterol diet').  
  
% -----  
% Rule to suggest a diet  
% -----  
suggest_diet(Disease) :-  
    disease_diet(Disease, Diet), !,  
    format("Suggested diet for ~w: ~w~n", [Disease, Diet]).  
suggest_diet(Disease) :-  
    format("No diet information available for ~w~n", [Disease]).
```

Output:

```
?- suggest_diet(diabetes).  
Suggested diet for diabetes: Low sugar, high fiber diet  
true.  
  
?- suggest_diet(asthma).  
No diet information available for asthma  
true.
```

6. Result

The Prolog program was successfully implemented to suggest an appropriate diet for a given disease, with a fallback message when no matching disease is found.

7. Conclusion

Thus, the disease-based dieting system was successfully implemented using Prolog. The experiment demonstrates a simple expert-system pattern: fact lookup with a cut for the success case and a fallback clause for the failure case.